

Weinberg-operator coefficient

$$T3 - D, \quad \alpha = -2 : \quad S_1 = (3, -1), \quad S_2 = (1, 0), \quad F = (2, -\frac{1}{2})$$

$$C_5 =$$

$$-\frac{2}{3} \left(-\frac{\sqrt{3}hM_F^3M_{S_2}^2\lambda_{T3}\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)} + \frac{\sqrt{3}hM_F^3M_{S_1}^2\lambda_{T3}\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)} - \frac{\sqrt{3}hM_F M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)} + \frac{\sqrt{3}hM_F M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)} \right)$$